

The Cyanobacteria Derived Toxin Beta-N-Methylamino-L-Alanine and Amyotrophic Lateral Sclerosis

There is mounting evidence to suggest that environmental factors play a major role in the development of neurodegenerative diseases like ALS (Amyotrophic Lateral Sclerosis). The non-protein amino acid beta-N-methylamino-L-alanine (BMAA) was first associated with the high incidence of Amyotrophic Lateral Sclerosis/Parkinsonism Dementia Complex (ALS/PDC) in Guam, and has been implicated as a potential environmental factor in ALS, Alzheimer's disease, and other neurodegenerative diseases. BMAA has a number of toxic effects on motor neurons including direct agonist action on NMDA and AMPA receptors, induction of oxidative stress, and depletion of glutathione. As a non-protein amino acid, there is also the strong possibility that BMAA could cause intraneuronal protein misfolding, the hallmark of neurodegeneration. While an animal model for BMAA-induced ALS is lacking, there is substantial evidence to support a link between this toxin and ALS. The ramifications of discovering an environmental trigger for ALS are enormous. In this article, we discuss the history, ecology, pharmacology and clinical ramifications of this ubiquitous, cyanobacteria-derived toxin.